

MOAZ HAMDY

Electrical MEP Design Engineer | Power, Lighting & Low Current | KNX / Smart Building

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Portfolio: moazhamdy-blip.github.io/portfolio

PROFESSIONAL SUMMARY

Electrical MEP design engineer with 3 years of experience across Egypt and Saudi Arabia, delivering power, lighting, low-current and KNX smart-building design for malls, towers, offices and villas. Prepares coordinated packages from early architectural and mechanical coordination through ETAP voltage-drop and short-circuit studies, SLDs, panel schedules, cable routing and low-current risers. Lighting studies in DIALux Evo; BIM workflow in Revit MEP and Navisworks. KNX Partner certified. Works to the Saudi Building Code (SBC), New Administrative Capital requirements and Civil Defense requirements.

CORE COMPETENCIES

- **Electrical:** power, small power, mechanical power, cable routing, SLD, panel schedules, earthing, EV charging, PV
- **Lighting:** DIALux Evo indoor, outdoor, facade; lux and uniformity analysis
- **Low current:** telecom, CCTV, fire alarm, parking, access control, disabled toilet alarm, sound system, risers
- **Smart building:** KNX design, automation shop drawings, ETS, GRMS
- **Analysis and BIM:** ETAP (voltage drop, short circuit), Revit MEP, Navisworks clash detection, AutoCAD

PROFESSIONAL EXPERIENCE

MEP Design Engineer

2025 – Present

ARCHRETE Engineering Consultancy | New Administrative Capital, Egypt

- Designed electrical packages (lighting, power, mechanical power, cable routing, earthing, EV, PV) for large New Administrative Capital malls including X1 Mall and OKASA Mall.
- Produced LV/MV single line diagrams, panel schedules and ETAP voltage-drop and IEC 60909 short-circuit studies.
- Prepared low-current documentation: telecom, CCTV, fire alarm, parking, disabled alarm, access control, sound system, cable routing and riser diagrams.
- Coordinated electrical room, shaft and service-space requirements with architectural and mechanical disciplines; compiled load calculation notes for the utility authority (OKASA: 5,656.1 kVA design load, three transformers, 1400 kW standby generator).

Pre-Sales Engineer

2025 (6 months)

Domtica Plus – Legrand Bticino & GVS distributor | Egypt

- Specified KNX and GRMS solutions for hospitality, villas and compounds (Marassi, Mountain View, Palm Hills, Mivida, Bellevie, Ras El Hekma).
- Prepared technical offers, BOQs and product comparisons for Legrand Bticino and GVS lines.

Electrical / MEP Design Engineer

2023 – 2025

ELEX, an Al Zamil Group company | Projects in Saudi Arabia

- Produced DIALux Evo lighting studies and reports for offices, villas, schools, a stadium, restaurant outdoor areas and hospitality facades (Ghiras Schools, Al Yamamah University Stadium, Al Bassam estate and others).
- Designed facade lighting for the Rosalie Hotel from CAD through lighting concept and DMX512 control system layout.
- Developed KNX automation shop drawings, KNX riser diagrams and lighting studies for Armada offices in Makkah and Madinah.
- Delivered electrical design for clients including the Ministry of Tourism, Al Bassam, Awtad Al-Fahd and Ramz Al-Bea'a, coordinating remotely with Saudi project and consultant teams.

SELECTED PROJECTS

- **X1 Mall / OKASA Mall (NAC):** full electrical and low-current design sets with ETAP studies, SLDs, panel schedules and risers.

- **Armada KNX, Makkah and Madinah:** KNX automation and lighting shop drawings with DIALux reports.
- **Rosalie Hotel facade:** facade lighting design, CAD to lit result.
- **Villa BIM project:** Revit electrical model, schedules, sheets, Navisworks clash detection and resolution.

TECHNICAL SKILLS

AutoCAD, Revit MEP, Navisworks, DIALux Evo, ETAP, ETS (KNX), Autodesk Construction Cloud, Microsoft Excel, Word, PowerPoint

CERTIFICATIONS AND TRAINING

KNX Partner: Basic, Advanced, Advanced HVAC. Revit MEP electrical modeling, documentation and coordination. Lighting design with DIALux Evo.

EDUCATION

B.Sc. Electrical Engineering, Shorouk Academy for Engineering and Technology

2023